

Day 5 : Assemble and Disassemble a PC

Task 1 : List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each.

Main Components Inside a Desktop PC

1. Motherboard :

- The main circuit board of the PC.
- It connects and allows communication between all major components.

2. CPU (Processor) :

- The CPU is the main processing unit.
- It performs calculations and executes instructions for running programs.

3. RAM (Memory) :

- RAM temporarily stores data and programs currently being used.
- More RAM helps the PC handle multiple tasks smoothly.

4. GPU (Graphics Card) :

- The GPU processes graphics and images for games, videos, and other visual applications.

5. HDD (Hard Disk Drive) :

- HDD is a storage device used to store the operating system, software, and personal files.

6. SSD (Solid State Drive) :

- SSD stores data like an HDD but uses flash memory.
- It provides faster boot and loading speeds.

7. Power Supply Unit (PSU) :

- The PSU converts electricity from the power outlet into suitable power for the PC components.

8. CPU Cooler (Fan & Heatsink) :

- The heatsink absorbs heat from the CPU, while the fan moves air to keep the CPU cool.

9. Case Fans :

- Case fans maintain airflow inside the cabinet by bringing in cool air and removing hot air.

10. Optical Drive :

- An optical drive is used to read or write CDs and DVDs, if the PC has one.

11. CMOS Battery:

- A small battery on the motherboard that helps maintain BIOS/UEFI settings and the system clock when the PC is powered off.

12. Expansion Cards :

- These are additional cards installed on the motherboard to provide features such as Wi-Fi, sound, or extra network connections.

Task 2 : Take clear photos or draw labeled diagrams of each step as you disassemble a desktop PC, and organize them in a document with captions explaining what you did at each step.

Desktop PC Disassembly – Steps

Step 1: Unplug the Power

Caption: I switched off the computer and unplugged the power cable from the power supply and wall socket. I also disconnected all external cables.

Step 2: Remove the Side Panel

Caption: I removed the screws holding the side panel and carefully opened the cabinet to access the internal components.

Step 3: Disconnect Internal Cables

Caption: I carefully disconnected the power and data cables connected to the motherboard, HDD/SSD, GPU, and other components.

Step 4: Remove RAM

Caption: I opened the RAM slot clips and carefully removed the RAM modules from the motherboard.

Step 5: Remove HDD/SSD

Caption: I disconnected the cables and removed the HDD/SSD from its mounting position inside the cabinet.

Step 6: Remove GPU (if installed)

Caption: I disconnected the GPU power cable, removed its mounting screw, released the PCIe slot clip, and carefully removed the graphics card.

Step 7: Remove CPU Cooler

Caption: I carefully disconnected the CPU fan cable and removed the CPU cooler from the motherboard.

Step 8: Remove Motherboard

Caption: I disconnected any remaining cables and removed the screws holding the motherboard to the cabinet. I then carefully took the motherboard out.

Step 9: Final Disassembled Components

Caption: After completing the disassembly, the main components were separated and identified, including the motherboard, CPU, RAM, storage drive, GPU, power supply, and cooling components.

Task 3 : Reassemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters.

PC Reassembly – Steps

Step 1: Install the Motherboard

I carefully placed the motherboard inside the cabinet and secured it with the correct screws. The motherboard is installed first because most other components connect to it.

Step 2: Install the CPU Cooler

I installed the CPU cooler on the CPU and connected its fan cable to the motherboard. Proper cooling is important before powering on the PC.

Step 3: Install RAM

I inserted the RAM modules into the correct RAM slots until the clips locked into place. RAM must be installed before other components that may block access to the slots.

Step 4: Install HDD/SSD

I placed the HDD/SSD in its mounting position and connected the required power and data cables. This allows the PC to access its storage.

Step 5: Install the GPU

I inserted the graphics card into the appropriate PCIe slot and secured it to the cabinet. I also connected its power cable if required.

Step 6: Connect Power Cables

I connected the motherboard power cable, CPU power cable, storage power cables, GPU power cable, and other required connections. These provide power to the installed components.

Step 7: Connect Front-Panel and Other Cables

I connected the power button, reset button, USB, audio, and other front-panel cables to the motherboard. This ensures that the cabinet controls and ports work correctly.

Step 8: Close the Cabinet

I checked that all components and cables were properly connected, then placed the side panel back and secured it with screws.

Step 9: Reconnect External Cables and Test

I connected the monitor, keyboard, mouse, network cable, and power cable. Finally, I turned on the PC to check that all components were working properly.

Why the Order Matters:

- The motherboard is installed first because most components connect to it.
- Components such as RAM, storage, and GPU are installed before closing the cabinet so their connections can be checked easily.
- Proper cable connections and final testing help prevent damage and ensure the PC starts correctly.

Task 4 : Create a checklist of safety precautions and tools needed before starting the PC assembly/disassembly process, and explain why each is important.

PC Assembly/Disassembly – Safety Checklist

Safety Precautions

- ✓ **Turn off the PC and unplug the power cable** – Prevents electric shock and protects components from electrical damage.
- ✓ **Disconnect all external cables** – Makes the PC safer and easier to handle.
- ✓ **Work on a clean, dry, and flat surface** – Prevents components from getting damaged or contaminated.
- ✓ **Use an anti-static wrist strap** – Helps prevent static electricity from damaging sensitive components.
- ✓ **Hold components by their edges** – Avoids touching pins, contacts, and circuits with your fingers.
- ✓ **Keep screws and small parts safely** – Prevents losing parts and makes reassembly easier.
- ✓ **Do not force any component** – Prevents damage to slots, connectors, and components.
- ✓ **Take photos before disconnecting cables** – Helps remember the correct cable connections during reassembly.

Tools Needed

- ✓ **Phillips-head screwdriver** – Used to remove and tighten most PC screws.
- ✓ **Anti-static wrist strap** – Protects components from electrostatic discharge (ESD).
- ✓ **Small container or magnetic tray** – Keeps screws and small parts organized.
- ✓ **Cable ties/Velcro straps** – Used to organize cables after assembly.
- ✓ **Compressed air or air blower** – Helps remove dust from the cabinet and components.
- ✓ **Thermal paste** – Used when reinstalling the CPU cooler to improve heat transfer between the CPU and heatsink.